The Egg, the Chicken, and the Big Bang

To see what this means, let's take the example of a pristine, low-entropy, fully formed egg. How did this low-entropy physical system come into being? Well, putting our trust back in memories and records, we all know the answer. The egg came from a chicken. And that chicken came from an egg, which came from a chicken, which came from an egg, and so on. But, as emphasized most forcefully by the English mathematician Roger Penrose,[15] this chicken-and-egg story actually teaches us something deep and leads somewhere definite.

A chicken, or any living being for that matter, is a physical system of astonishingly high order. Where does this organization come from and how is it sustained? A chicken stays alive, and in particular, stays alive long enough to produce eggs, by eating and breathing. Food and oxygen provide the raw materials from which living beings extract the energy they require. But there is a critical feature of this energy that must be emphasized if we are to really understand what's going on. Over the course of its life, a chicken that stays fit takes in just about as much energy in the form of food as it gives back to the environment, mostly in the form of heat and other waste generated by its metabolic processes and daily activities. If there weren't such a balance of energy-in and energy-out, the chicken would get increasingly hefty.

The essential point, though, is that all forms of energy are not equal. The energy a chicken gives off to the environment in the form of heat is highly disordered—it often results in some air molecules here or there jostling around a touch more quickly than they otherwise would. Such energy has high entropy—it is diffuse and intermingled with the environment—and so cannot easily be harnessed for any useful purpose. To the contrary, the energy the chicken takes in from its feed has low entropy and is readily harnessed for important life-sustaining activities. So the chicken, and every life form in fact, is a conduit for taking in low-entropy energy and giving off high-entropy energy.

This realization pushes the question of where the low entropy of an egg originates one step further back. How is it that the chicken's energy source, the food, has such low entropy? How do we explain this aberrant source of order? If the food is of animal origin, we are led back to the